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Research Article

Characterization of PEG in monomethyl ethers of PEG by 2-D LC

PEGs present in samples of the corresponding monomethyl ethers are analyzed by 2-D LC with LC at critical conditions as the first and liquid adsorption chromatography as the second dimension. The fractions from the first dimension are transferred to the second one by heart-cutting using a ten-port valve. The efficiency of the separation is dramatically improved by adding water to the eluate from the first dimension before the switching valve in order to have a weaker eluent in the fraction than that of the second dimension. By this procedure, the fractions are focussed and reconcentrated. The molar mass distribution of the PEG fraction can be determined with good accuracy even at very low amounts of PEG in the PEG-monomethyl ethers (down to 0.5%).

Keywords: 2-D HPLC / Critical conditions / ELSD / LC / PEG
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1 Introduction

Monofunctional polyethylene oxides are used in many different fields. While the monoalkylethers of PEG with longer alkyl end groups are used as non-ionic surfactants (such as fatty alcohol ethoxylates), the monomethyl ethers (PEG-MME) can be used to produce amphiphilic block and graft copolymers, where the oxyethylene (EO) chains are the hydrophilic component, while the other block or the main chain is hydrophobic.

Other important applications are the surface modification of polymers and the PEGylation of proteins, drugs, *etc.*, which leads to products with different physical and biological properties, such as increased water solubility, reduced degradability by proteolytic enzymes, *etc.* [1–7]. Graft polymers are produced by copolymerization of vinyl monomers with macromonomers, which are obtained by reaction of PEG-MME with methacryloyl chloride [8]. Obviously, traces of PEG in the PEG-MME will lead to dimethacrylates, which cause crosslinking. In this case, hydrogels will be obtained instead of soluble graft copolymers.

A similar problem may occur in the modification of proteins with PEG-MME. Presence of PEG having two active sites results in cross-linking of two proteins or even the formation of aggregates [9].

The quality of all these products obviously depends on the purity of the PEG-MME. Monoalkyl ethers of PEG are synthesized by (typically anionic) polymerization of ethylene oxide with methanol or the MME of triethylene glycol [8] as initiator and sodium hydroxide or a similar catalyst (such as potassium naphthalene [10]). If the polymerization mixture contains traces of water, PEGs may be formed as side product. As PEGs can grow in both directions, their molar mass will typically be higher than that of the MMEs (in general approximately twice as high).

Special polymerization conditions have been described, which allow the synthesis of (almost) pure MMEs (without diols) [8, 10]. The purification of PEG-MME by removal of PEG has also been described [9, 11]. Lapienis *et al.* [11] purified PEG-MME by preparative LC on silica gel. Selisko *et al.* [9] applied preparative vesicle chromatography and (SEC) on Superose.

The question is, however, the reliability of the methods for their analysis. In the literature this has been done by size exclusion chromatography (SEC), NMR, and matrix-assisted laser desorption ionization time-of flight mass spectroscopy (MALDI-TOF-MS) [8, 10].

All techniques for such analysis have, however, some limitations. For example gas chromatography can be used only for separation of low molecular weight oligomers, NMR yields only average values, moreover it is not sensitive enough for the determination of low PEG contents [9]. MALDI-TOF-MS can perfectly resolve all oligomers, but is problematic with respect to quantitation, especially in the low molecular range (*i.e.* close to the molar mass of the matrix).

In principle, a separation of polymers according to functionality can be achieved by LC at critical conditions (LCCC) [8, 12–15].

At a special mobile phase composition and temperature (the critical point of adsorption, CAP) the retention of a polymer chain is independent of molar mass, and separation according to functionality becomes possible. One may expect that at the CAP for PEG on a reversed phase column

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Abbreviations: CAP, critical point of adsorption; **LAC,** liquid adsorption chromatography; **LCCC,** LC at critical conditions; **MME,** monomethyl ether the methoxy end group of PEG-MME may be slightly adsorbed, which could be utilized to separate PEG and PEG-MME. Indeed, such a separation has been described recently [16].

Besides the total amount of the by-product PEG (which can be obtained from LCCC), its molar mass distribution (MMD) is important, as the molar concentration of diols depends on the total amount of the PEG and its number average of molar mass. In principle, this may be achieved by MALDI-TOF-MS or by a 2-D separation, which can follow different strategies: LCCC in the first dimension with heart-cutting and liquid adsorption chromatography (LAC) as the second dimension or comprehensive chromatography with LAC as the first dimension and LCCC as the second dimension.

In this study, we wanted to find conditions, under which a full resolution of all oligomers can be achieved, and if this is possible, which amounts of PEG could be identified and quantitatively determined.

2 Scope and limitations of different strategies in 2-D HPLC

In principle, different approaches can be applied in 2-D separations:

The off-line approach uses a fraction collector to collect fractions, which are subsequently injected into the second dimension. This approach has several advantages: If the mobile phase used in the second dimension is not compatible with that in the first one, the mobile phase from the first dimension can be removed (*e.g.* by freeze drying) and the sample subsequently dissolved in the mobile phase used in the second dimension. Moreover, it allows repeated analyses under different conditions and by different (independent) methods, *e.g.* by spectroscopic techniques, such as MALDI-TOF-MS [17]. If the samples can be injected directly, automated operation is possible. The disadvantage is that it is laborious and has to be run on a larger column, which may have a lower efficiency. Moreover, the fractions may be contaminated on handling.

The on-line approach does not require any handling of fractions: the eluate from the first dimension is transferred to the second dimension *via* a switching valve [18]. On-line coupling can only be applied if the mobile phases in both dimensions are compatible. Basically, there are two different approaches: heart-cutting and comprehensive chromatography.

In heart-cutting, single fractions from the first dimension are transferred to the second dimension. If necessary, they may be stored in a short column on which they are completely adsorbed. If the second dimension runs in a stronger eluent, the fraction is reconcentrated. This works well if sufficiently resolved peaks are obtained in the first dimension.

In comprehensive chromatography, predefined volumes of the eluate from an entire chromatogram in the first dimension are automatically injected into the second dimension. Obviously, a fraction from the first dimension can only be injected in the second dimension when the previous one has been completely analyzed. Consequently, the main problem in comprehensive chromatography is the mismatch of flow rates, the sizes of the sample loops and the columns. An entire chromatogram in the second dimension has to be finished in the same time, which is required to fill the sample loop in the first dimension. Hence, the first dimension has to be run at a very low flow rate (far from the van Deemter optimum), and the second one at a very high one (which may cause problems with the back pressure).

For the separation of PEG and PEG-MME there can be three different strategies: In comprehensive LC, the first dimension (which may be slow) should be LAC and the second one (which must be finished after a defined elution volume) must be LCCC. It is, however, not very likely, that in the second dimension a separation of PEG and PEG-MME can be achieved on a short column if a quite large volume is injected.

In heart-cutting, the order of the dimensions does not matter, since there is no time limit in both dimensions. For practical reasons it is reasonable to run LCCC in the first dimension to yield just two fractions, which are subsequently analyzed by LAC in the second dimension. For comparison, we have tried to apply all the approaches mentioned above. As will be shown, on-line 2-D LC with heart-cutting proved to be superior over the other approaches.

3 Materials and methods

These investigations were performed using a gradient system Ultimate 3000, consisting of a pump DGP-3600A, solvent degasser SRD-3600, column thermostat TCC-3000, autosampler WPS-3000SL, all from Dionex (Germerink, Germany), and an evaporative light scattering detector (ELSD) PL-ELS 2100 (Polymer Laboratories/Varian, Church Stretton, Shropshire, UK). All separations were performed at a flow rate of 0.5 mL/min. The mobile phases were mixed by the gradient pump, so their compositions are given in volume percent. Data acquisition and processing was performed using the software Chromeleon (Dionex).

The following columns were used in this study (specifications given by the producer):

Synergi 4 μ MAX-RP 80A (Phenomenex, Torrance, CA, USA): silica-based dodecyl phase; 250 × 4.6 mm; particle diameter = 4 μm; pore size = 80 Å, surface area: 475 m^2/g, carbon load: 17%.
Synergi Fusion RP: silica-based dodecyl phase; 250 × 4.6 mm; particle diameter = 4 μm; nominal pore size = 80 Å (Phenomenex).

The solvents (methanol, acetone, and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany).

PEG and it MMEs (Me-PEG) were purchased from Fluka (Buchs, Switzerland).

4 Results and discussion

First of all, we wanted to reproduce the LCCC separation described by Barman [16] with our equipment. PEG 600 and PEG-MME 350 can be reasonably separated on the Synergi Fusion RP column in 82% methanol–water at 25°C. A much better resolution could be achieved on the same column in 32 vol% acetone–water at the same temperature (Fig. 1).

In PEG 350 one may expect a PEG similar to PEG 600, in PEG-MME 550 similar to PEG 900. In both cases, an excellent separation could be achieved.

Evidently, the samples used in this study were fairly clean, which was also confirmed by MALDI-TOF-MS. Hence, we used the pure samples as well as mixtures of PEG-MME and PEG (in a different ratio) for the further investigations.

As has been mentioned above, 2-D separation could be possible in different ways: off-line (with preparative separation by LCCC in the first dimension) or on-line.

In principle, the off-line separation would be reasonable, but in practice no sufficient resolution could be achieved on a semipreparative column.

In comprehensive chromatography, there are several possibilities, all of which are problematic: as a separation in the second dimension must be finished very rapidly, this could be SEC or LCCC, as LAC cannot be finished in 1–2 min. On the other hand, no LCCC separation of PEG and PEG-MME is possible on a short column (10 cm or less), when the injected volume is 100 μL or more.

The most promising solution appeared to be LCCC in the first dimension (as shown above) with heart-cutting of the PEG-MME and LAC in the second dimension. The setup was the same as usual in comprehensive LC.

As the first task, the correct switching points had to be found. This was achieved by the following procedure. Both pumps were operated in the same mobile phase, which corresponds to the CAP for EO (32% acetone). Each chromatogram was started with the ten-port valve in the 1_2 position. At the selected time the content of the sample loop was injected in the second dimension by switching the valve to the 10_1 position, and at the end of the chromatogram back to the 1_2 position.

The efficiency of this procedure was tested by running the second dimension also at the CAP for PEG. We analyzed a sample consisting of PEG-MME 350 and PEG 600 (50% each) and found that there was always an overlap of both fractions, as can be seen in Fig. 2. This was not a big surprise: when a zone of 200 μL was injected in the second dimension, this caused rather broad peaks. Hence, we tried to focus the fractions from the first dimension. For this purpose, we used an approach similar to the one described in previous papers [19, 20].

Using a third pump, we added water to the eluate from the first dimension before the switching valve (see Fig. 3) in order to make the mobile phase weaker than the one used in the second dimension. When now 200 μL of a solution with 16% acetone was injected on a column running in 32% acetone, the fractions were strongly adsorbed and subsequently eluted as a narrow zone by the stronger eluent, as can be seen in Fig. 4.

This works very well, if the mobile phase in the second dimension corresponds to critical conditions, but when we tried to separate the PEG fraction by LAC with a gradient from 13 to 20% acetone in 20 min, the peaks of the lower oligomers were very broad (Fig. 5). Only the peaks of higher oligomers (which already eluted in a much stronger eluent) were reasonably narrow.

Hence, we increased the side flow from 0.5 to 1.0 mL/min, which resulted in a sufficient difference in solvent strength, and obtained perfectly resolved peaks as well for the PEG as also for the PEG-MME fraction (Fig. 6).

Obviously, the switching times had to be adjusted to the higher flow rate in the sample loop. Again we used a mobile phase corresponding to the CAP in the second dimension. Under such conditions, the peaks were strongly focused on the column of the second dimension. When a mixture of the

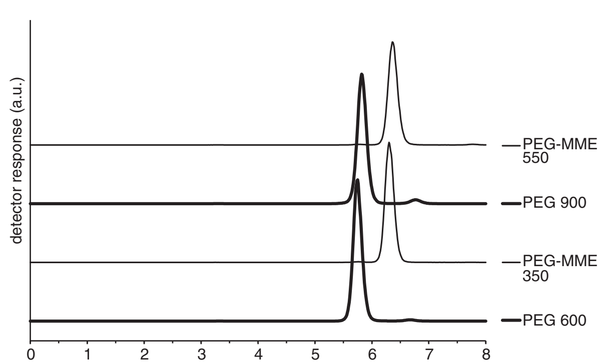

Figure 1. LCCC of different PEG and PEG-MME samples on the Synergi Fusion RP column in 32% acetone–water.

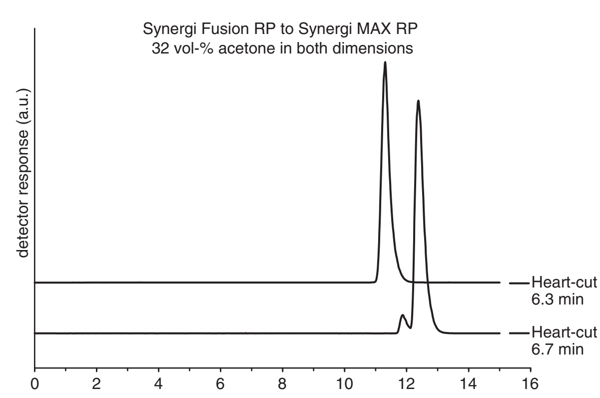

Figure 2. LCCC × LCCC of PEG 600 and PEG-MME 350. 1st dimension: Synergi Fusion RP, 2nd dimension Synergi MAX RP, both in 32% acetone–water.

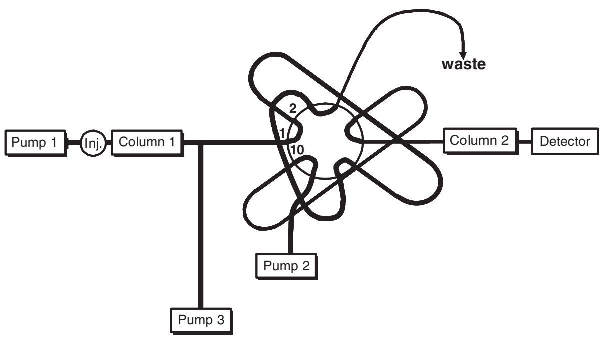

Figure 3. Experimental setup for 2-D separations with addition of water behind the column used in the first dimension.

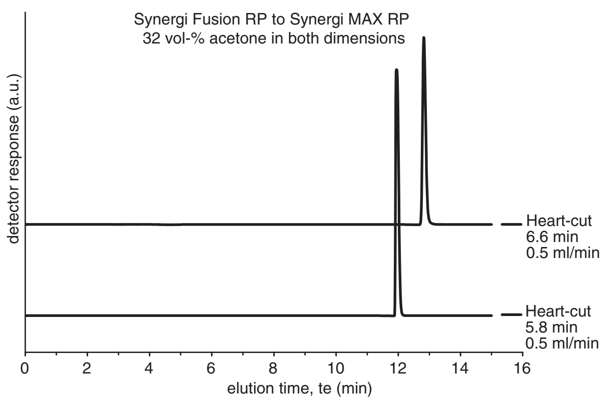

Figure 4. LCCC × LCCC of PEG 600 and PEG-MME 350. Conditions were as in Fig. 2, but a side flow of 0.5 mL/min. Top: PEG-MME, bottom: PEG.

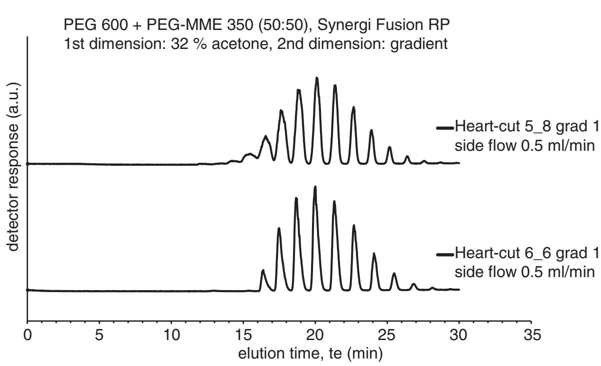

Figure 5. LCCC × LAC of a mixture of PEG 600 and PEG-MME 350. 1st dimension: Synergi Fusion RP, 32% acetone–water, 2nd dimension: Synergi MAX RP, gradient from 13 to 20% acetone in 20 min. Side flow 0.5 mL/min. Top: PEG, bottom: PEG-MME.

same amounts of PEG 600 and PEG-MME 350 was analyzed with optimized switching times, pure fractions were obtained, which did not contain the other homologous series (see Fig. 7).

As the next step, we optimized the gradient profile. A steeper gradient produced an even better peak pattern.

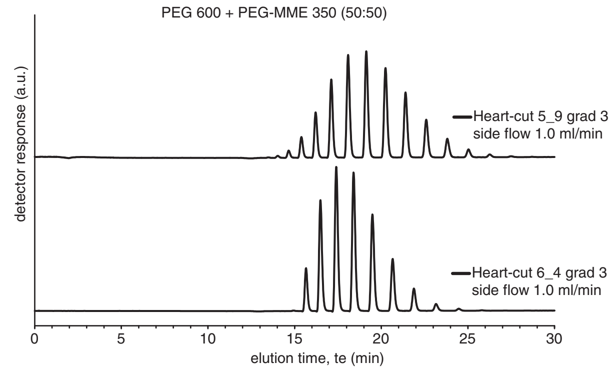

Figure 6. LCCC × LAC of a mixture of PEG 600 and PEG-MME 350. 1st dimension: Synergi Fusion RP, 32% acetone–water, 2nd dimension Synergi MAX RP, gradient from 15 to 20% acetone in 20 min. Side flow 1.0 mL/min. Top: PEG, bottom: PEG-MME.

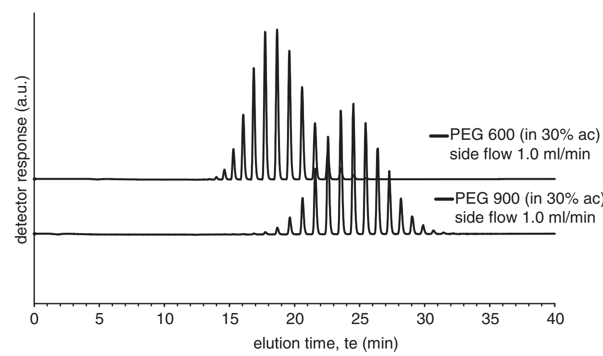

Figure 7. LCCC × LCCC of PEG 600, PEG-MME 350 and a 50:50 mixture of both. Conditions were as in Fig. 2, but with a side flow of 1.0 mL/min. Switching time for PEG fraction (5.9 min).

When the gradient was run up to a higher acetone concentration (15–25% acetone in 30 min), it was even possible to separate PEG 900 and PEG-MME 550 with a baseline resolution of all oligomers. The chromatograms obtained with the pure PEGs are shown in Fig. 8, and those obtained with mixtures of the PEG-MME with 10% of the corresponding PEG are shown in Fig. 9. Evidently, the results agree very well.

The individual peaks were identified using monodisperse oligomers such as hexa or octa-EG as external or internal standards (Fig. 10).

Finally, we wanted to find out the lowest concentration of PEG in an MME sample, which would allow a determination of not only its total amount but also its MMD. For this purpose we prepared test mixtures (by mixing stock solutions with 5–6 g/L) with a content of PEG in the range from 0.5 to 50 wt%. The results thus obtained with mixtures containing 0.5–5% PEG are shown in Fig. 11. Even for a mixture with 0.5% PEG the agreement with pure PEG was perfect.

The molar mass distributions of PEG 600 and those obtained for the PEG fractions in the mixtures shown above

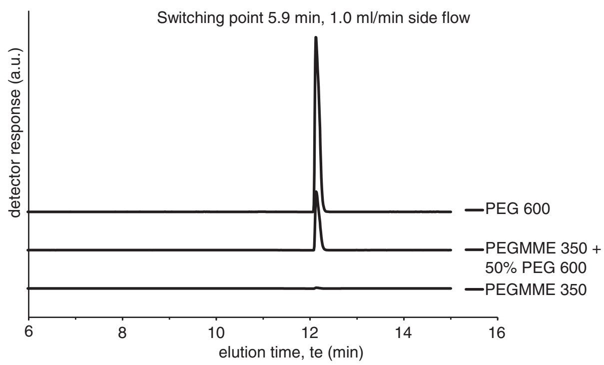

Figure 8. LCCC $\times$ LAC of PEG 600 and PEG 900. Conditions were as in Fig. 5. Switching time for PEG fraction (5.9 min).

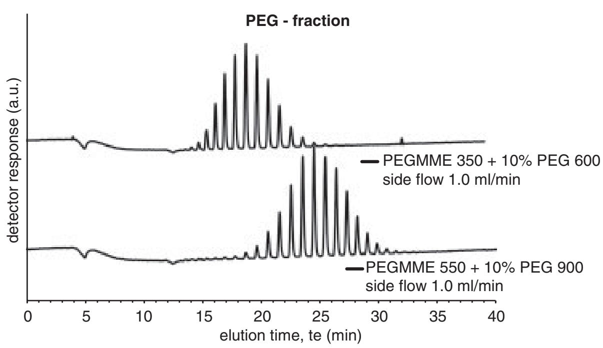

Figure 9. LCCC $\times$ LAC of mixtures of PEG and PEG-MME. Conditions were as in Fig. 6. Switching time for PEG fraction (5.9 min).

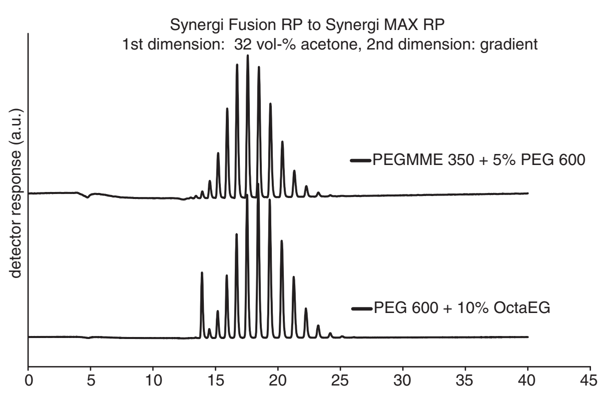

Figure 10. LCCC $\times$ LAC of PEG 600 (spiked with octa-EG) and a mixture of PEG-MME 350 and 5% PEG 600. Conditions were as in Fig. 6.

are shown in Fig. 12, the corresponding molar mass averages are listed in Table 1. Evidently, the accuracy of the results is very good. The agreement of the molar mass averages is excellent for PEG contents of 5% and larger, and still reasonably good below 1%.

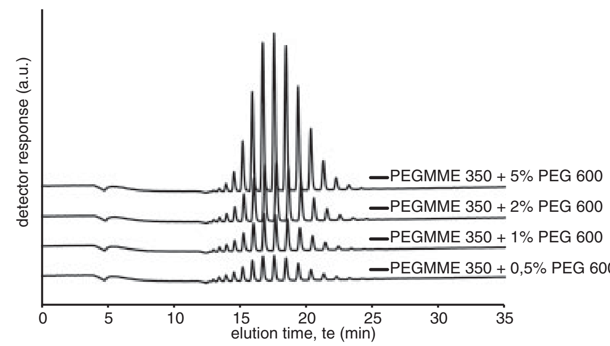

Figure 11. LCCC $\times$ LAC of the PEG-MME fraction in mixtures of PEG and PEG-MME. Conditions were as in Fig. 6. Switching time for PEG fraction (5.9 min).

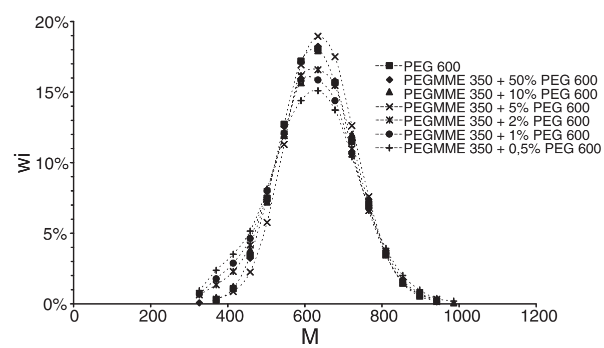

Figure 12. MMD of PEG 600 and the PEG fraction in mixtures of PEG-MME with different amounts of PEG 600, as obtained by LCCC $\times$ LAC (conditions were as in Fig. 6).

Table 1. Molar mass averages (weight average M_w and number average M_n) of PEG 600 (from LAC) and of the PEG fraction in mixtures of PEG 600 and PEG-MME 350, as obtained by 2-D HPLC with heart-cutting

	PEG 600	PEG-MME 350+PEG 600					
		50% PEG	10% PEG	5% PEG	2% PEG	1% PEG	0.5% PEG
M_w	634	635	637	643	626	622	620
M_n	619	620	621	629	606	601	596
M_w/M_n	1024	1024	1026	1022	1032	1036	1041

5 Concluding remarks

In the analysis of PEG-MME traces of PEG can easily be determined by LCCC. The molar mass distributions of these impurities can be determined even at very low concentrations of PEG by 2-D chromatography using LCCC as the first and LAC as the second dimension. Heart-cutting proved to be the method of choice for the transfer of fractions from the first to the second dimension. The fractions can be focused by adding water to the eluate after the first dimension to yield a composition with a lower solvent strength than that of the mobile phase in the second dimension.

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The authors have declared no conflict of interest.

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